Name: _____ Class: _____ No.: _____

Antibiotics（抗生素） and bacterial growth

Study sequence: explanation → worked example → guided practice → independent application → checking and correction.

All plates and data are teaching simulations. Do not operate the VL or culture MRSA（耐甲氧西林金黃葡萄球菌） physically. Shared simulated conditions: 30°C and 24 hours.

The history of antibiotic（抗生素） discovery—an unexpected observation



In 1928, Fleming noticed contamination by Penicillium（青黴菌） on plates growing staphylococci, with reduced bacterial growth nearby. He investigated an antibacterial substance produced by this fungus. Later, Florey, Chain and their team advanced the therapeutic use of penicillin（青黴素）, an antibiotic（抗生素）.

Note: Fleming’s historical work involved staphylococci, not MRSA（耐甲氧西林金黃葡萄球菌）; MRSA（耐甲氧西林金黃葡萄球菌） was first formally reported in 1961.

Image source: https://www.cbsnews.com/news/almanac-the-discovery-of-penicillin（青黴素）/

Research context

A wound infection has not clearly improved after initial antibiotic（抗生素） treatment; testing identifies MRSA（耐甲氧西林金黃葡萄球菌）. This worksheet uses static images from three independent plates to compare X, Y, Z and the matching carrier control C for visible bacterial growth.

Under the stated simulation conditions, how do X, Y and Z affect visible growth of these MRSA（耐甲氧西林金黃葡萄球菌） bacteria?

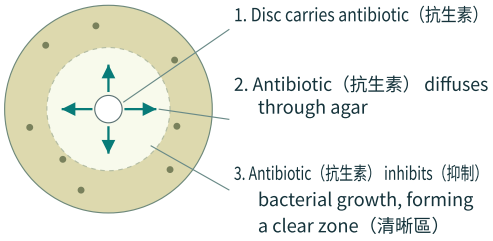
Learn the core concepts first

- Antibiotics（抗生素） mainly treat bacterial infections, not viral colds or influenza.
- Antibiotic（抗生素） diffuses outward from the disc, usually decreasing in concentration with distance. A clear zone（清晰區） forms where growth is still inhibited（抑制）. More susceptible（對抗生素敏感的） bacteria may be inhibited（抑制） at a lower concentration, allowing a larger zone.
- A clear zone（清晰區） has no visible bacterial growth; it does not prove all bacteria died. Diffusion and preparation also affect size, so the largest zone is not the best patient treatment.
- MRSA（耐甲氧西林金黃葡萄球菌） resists methicillin（甲氧西林）, not necessarily every antibiotic（抗生素）. No zone is not a clinical resistance（抗藥性） classification: this model has no clinical breakpoints and cannot be generalised to all MRSA（耐甲氧西林金黃葡萄球菌）.
- If a few resistant（具抗藥性的） bacteria already exist, antibiotic（抗生素） selection pressure（選擇壓力） favours their survival, reproduction（繁殖） and inheritance（遺傳） of resistance traits（抗藥性特徵）. Their proportion may rise and effectiveness may decline without their occupying the entire population.
- Bacteria do not purposefully produce the resistance（抗藥性） they need to adapt, and patients themselves do not develop bacterial resistance（抗藥性）. Misuse（不當使用） or overuse（過度使用） adds unnecessary selection pressure（選擇壓力）.

Complete B yourself and preserve your first answers before reading C. The examples in A are not the results for X/Y/Z/C.

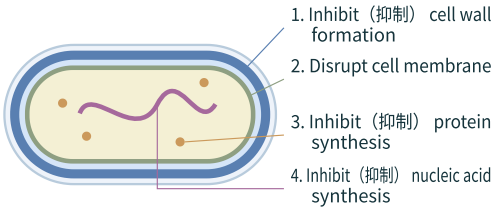
Worked examples: applying concepts to investigation and measurement

A. Experimental principle: antibiotic（抗生素） diffusion and clear-zone（清晰區） formation



Schematic, not to scale; it does not prove complete killing.

B. Structures and possible action sites



The pale outer layer represents a capsule (not present in all bacteria); blue is the cell wall, green is the cell membrane, and the purple strand is DNA. These mechanisms are not proved by this disc experiment.

Testable hypotheses and variables

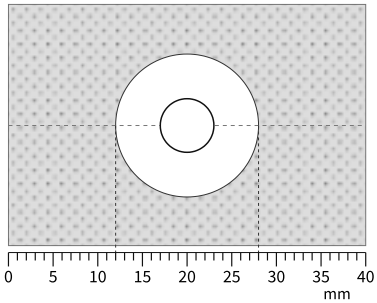
Example: compare sample P with the same carrier control for bacterium T. “With other conditions equal, P has a larger external-zone total diameter than the carrier control.” This names a comparison and a measurable outcome, but cannot prove complete killing from zone size. It is not an X/Y/Z prediction.

Example: if only disc size changes for the same sample, disc size is independent; external-zone presence and total diameter are dependent. Keep bacteria and carrier alike. For the X/Y/Z task, classify the factor actually being changed.

Control, spacing and measurement

C contains the same carrier without antibiotic（抗生素） as a comparison baseline. Call it a water control only when the carrier is water. Space discs apart and away from the edge to avoid overlap and difficult interpretation.

Example S (not X/Y/Z/C): an external zone surrounds a 6 mm disc. Its ends read 12 mm and 28 mm; total diameter = 28−12 = 16 mm, including the disc. With only a disc and no external zone, this model records 0, not 6.



Mean total diameter calculation

Example S has 12, 16 and 20 mm from three independent plates: mean = (12+16+20) ÷ 3 = 16.0 mm. Repeats check variation. First investigate spreading, contamination, disc positions and conditions, then consider additional independent tests. Never delete data for disagreeing with a hypothesis. Three is a classroom arrangement, not the only reliable number.

Complete B yourself and preserve your first answers before reading C. The examples in A are not the results for X/Y/Z/C.

Guided practice

Circle applicable options; answer written questions with one or two sentences or key terms.

Refer to the historical image in A. Describe visible bacterial growth near the fungus, separating observation from an explanation of its cause.

Compare two claims: A, “This sample must kill all MRSA（耐甲氧西林金黃葡萄球菌）”; B, “Under the same conditions, the total clear-zone（清晰區） diameter around a disc with sample M is greater than around the carrier control.” Circle the more testable claim (A/B), and explain why clear-zone（清晰區） data can test it.

Hint: state the comparison and an observable outcome. Zone data cannot directly prove “all killed”.

Another context: compare unknown sample U with the same carrier control using a different bacterium. Write a hypothesis testable with external-zone presence and total diameter; do not make a new prediction about X/Y/Z.

What is C? Explain its comparison baseline and why the remaining conditions must be the same.

(Hint: The control group provides a baseline for comparison. It helps check whether changes in the experiment are caused by the independent variable rather than other external factors. Therefore, the control group does not receive the treatment being tested.)

Identify the variables

Tick all applicable boxes; more than one may be selected.

Independent variable（The factor you change）

- ☐ Disc size
- ☐ Sample in the disc
- ☐ Composition of the agar plate（瓊脂板）
- ☐ Visible clear zone（清晰區）and its total diameter
- ☐ Type of bacteria used
- ☐ Initial bacterial distribution
- ☐ Preset sample preparation

Dependent variable（The outcome you observe/measure）

- ☐ Disc size
- ☐ Sample in the disc
- ☐ Composition of the agar plate（瓊脂板）
- ☐ Visible clear zone（清晰區）and its total diameter
- ☐ Type of bacteria used
- ☐ Initial bacterial distribution
- ☐ Preset sample preparation

Controlled variables（The factors kept the same）

- ☐ Disc size
- ☐ Sample in the disc
- ☐ Composition of the agar plate（瓊脂板）
- ☐ Visible clear zone（清晰區）and its total diameter
- ☐ Type of bacteria used
- ☐ Initial bacterial distribution
- ☐ Preset sample preparation

What are the assumptions of this investigation?

An assumption is a condition we have not proved but take as true, to help make the experiment results valid.

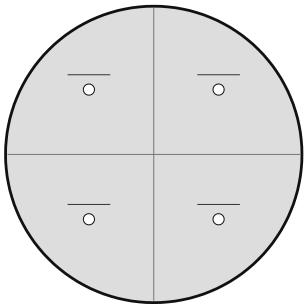
Select all applicable assumptions; you may choose more than one.

- ☐ Bacteria are initially distributed evenly on each agar plate（瓊脂板）.
- ☐ Equipment does not introduce contamination.
- ☐ All plates use the same simulated observation conditions.
- ☐ If no colonies（菌落） form near a disc, this means all bacteria at that location have died.

Complete labels and check fair comparison

Label the four discs X, Y, Z and C, once each. Explain why discs need space between them and from the edge. Label order does not affect credit.

You do not need to draw every detail carefully. A simple drawing with clear labels is enough.



Why are spacing and edge distance needed?

Circle flawed arrangements (more than one): A, “Pick up the next disc without re-sterilising the forceps”; B, “Use another bacterium on one plate, but combine all plates directly for comparison”; C, “Use the same carrier, bacteria and observation conditions”. Identify contamination or unfair comparison.

Plan A: remeasure the same plate three times. Plan B: independently prepare and observe three new plates once each. Which provides independent repeats, and how do they help check variation?

This lesson uses three plates. Why is “three” not the only correct number for every study? What feasibility limits matter when comparing three with more independent plates?

Diagram reading and procedure ordering (not practical performance)

These cards describe a simulated procedure. Order the letters logically; prepare three plates independently before the shared observation. Answering does not demonstrate sterilisation（滅菌／消毒）, spreading or incubation（培養） performance.

P. Observe all three prepared new plates under shared simulated conditions of 30°C and 24 hours.

Q. Spread vertically with a sterile（無菌的） spreader（塗布器） and rotate for even distribution.

R. Divide and label the plate bottom X/Y/Z/C.

S. Cover and invert; repeat the preparation independently on new plates.

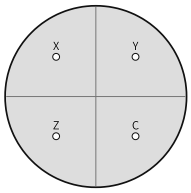
T. Before each disc, sterilise（滅菌／消毒） forceps with the alcohol beaker and lamp, then place it at the matching position.

U. Release diluted（已稀釋的） bacterial suspension（細菌懸液） at the top centre.

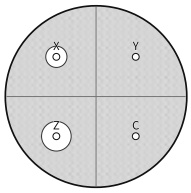
Circle errors (more than one): A, “Place discs before spreading evenly”; B, “Call remeasurement of one zone a new plate’s independent result”; C, “Retain differing readings from three independent plates”. State what should be checked.

Independent application: observe, measure and interpret • Plate 1

All plates and data are teaching simulations. Do not operate the VL or culture MRSA（耐甲氧西林金黃葡萄球菌） physically. Shared simulated conditions: 30°C and 24 hours.



Before simulated incubation（培養） (no visible growth result yet)

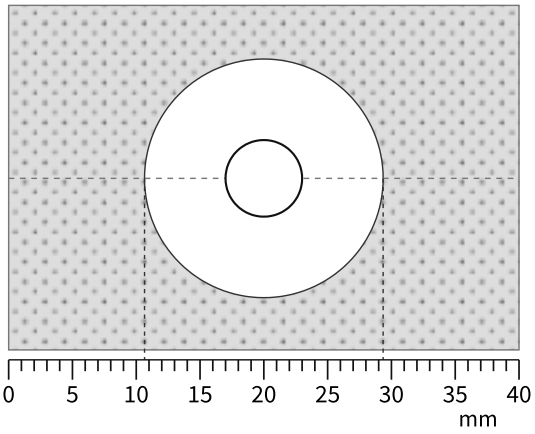


After simulated incubation（培養）: 30°C, 24 hours

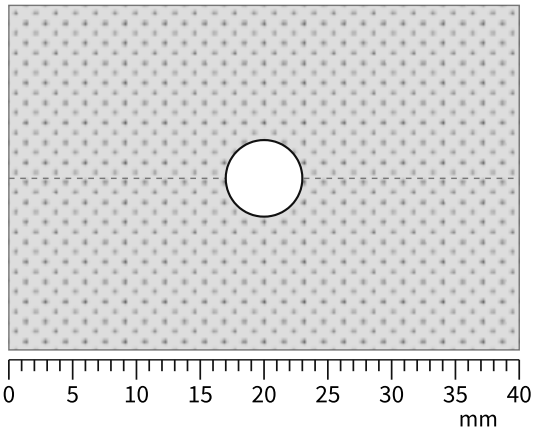
Read both ends on the illustrated mm scale and subtract for the total diameter. Use the labelled scale, not the physical length of the printed image.

When a zone extends beyond the disc, the total diameter includes the 6 mm disc. Without an external zone, record 0; the 6 mm disc alone is not a clear zone（清晰區）.

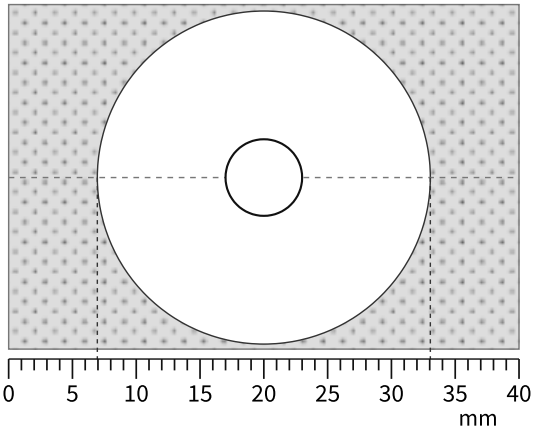
X



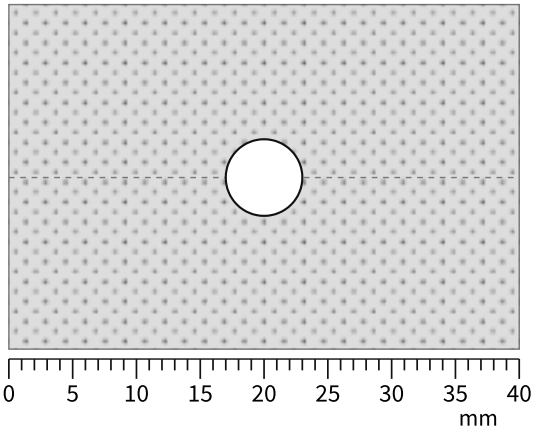
Y



Z



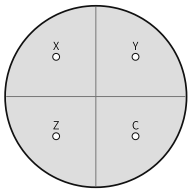
C (control)



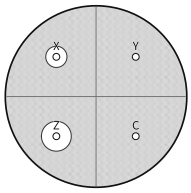
Sample	Visible zone outside the disc?	Total clear-zone（清晰區） diameter (mm)
X	☐ Yes ☐ No	
Y	☐ Yes ☐ No	
Z	☐ Yes ☐ No	
C (control)	☐ Yes ☐ No	

Independent application: observe, measure and interpret • Plate 2

All plates and data are teaching simulations. Do not operate the VL or culture MRSA（耐甲氧西林金黃葡萄球菌） physically. Shared simulated conditions: 30°C and 24 hours.



Before simulated incubation（培養） (no visible growth result yet)

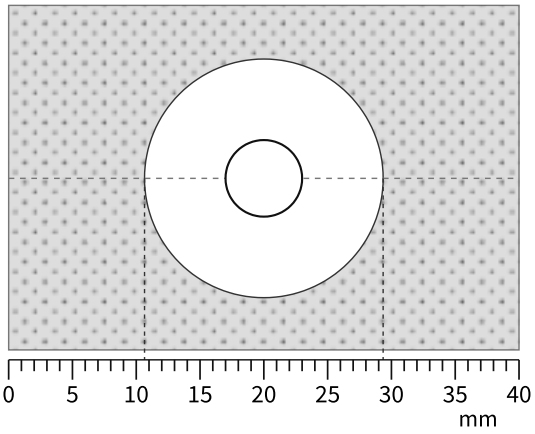


After simulated incubation（培養）: 30°C, 24 hours

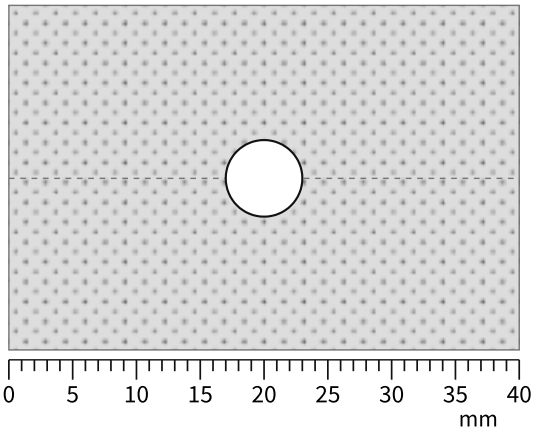
Read both ends on the illustrated mm scale and subtract for the total diameter. Use the labelled scale, not the physical length of the printed image.

When a zone extends beyond the disc, the total diameter includes the 6 mm disc. Without an external zone, record 0; the 6 mm disc alone is not a clear zone（清晰區）.

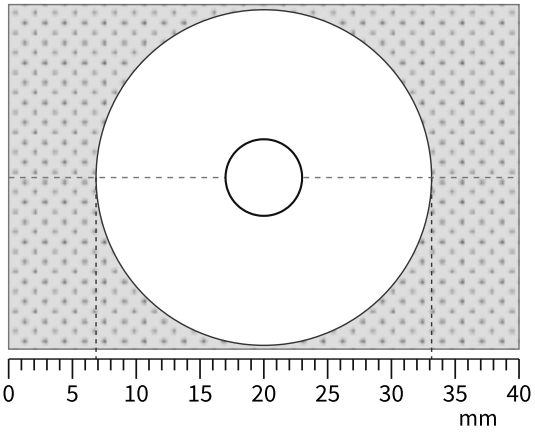
X



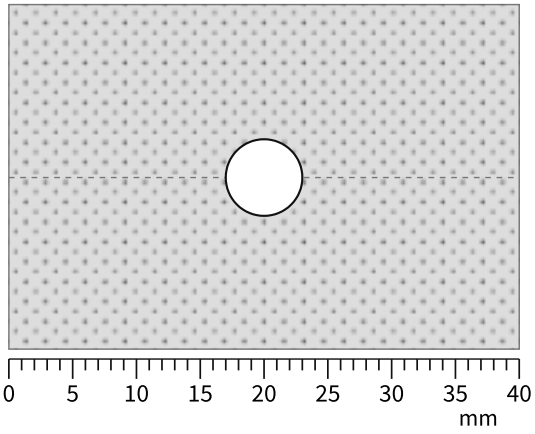
Y



Z



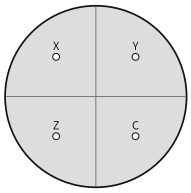
C (control)



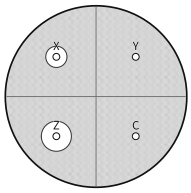
Sample	Visible zone outside the disc?	Total clear-zone（清晰區） diameter (mm)
X	☐ Yes ☐ No	
Y	☐ Yes ☐ No	
Z	☐ Yes ☐ No	
C (control)	☐ Yes ☐ No	

Independent application: observe, measure and interpret • Plate 3

All plates and data are teaching simulations. Do not operate the VL or culture MRSA（耐甲氧西林金黃葡萄球菌） physically. Shared simulated conditions: 30°C and 24 hours.



Before simulated incubation（培養） (no visible growth result yet)

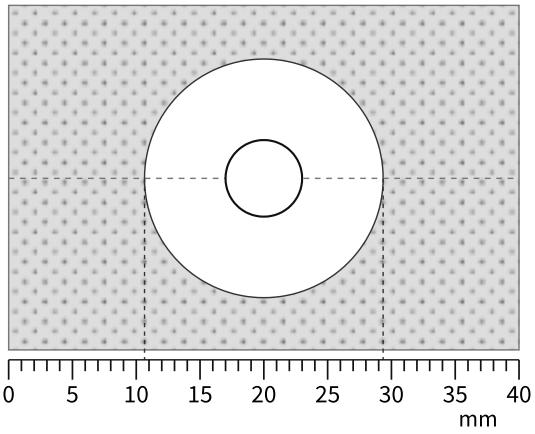


After simulated incubation（培養）: 30°C, 24 hours

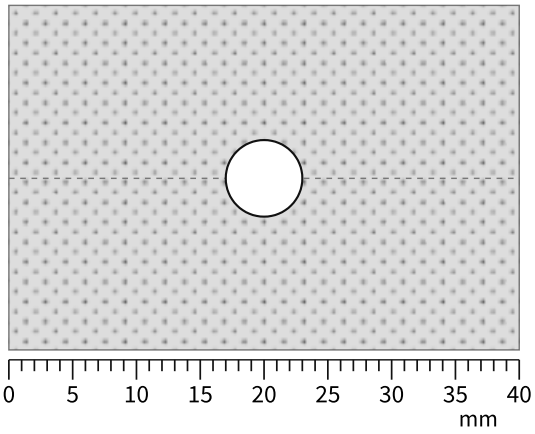
Read both ends on the illustrated mm scale and subtract for the total diameter. Use the labelled scale, not the physical length of the printed image.

When a zone extends beyond the disc, the total diameter includes the 6 mm disc. Without an external zone, record 0; the 6 mm disc alone is not a clear zone（清晰區）.

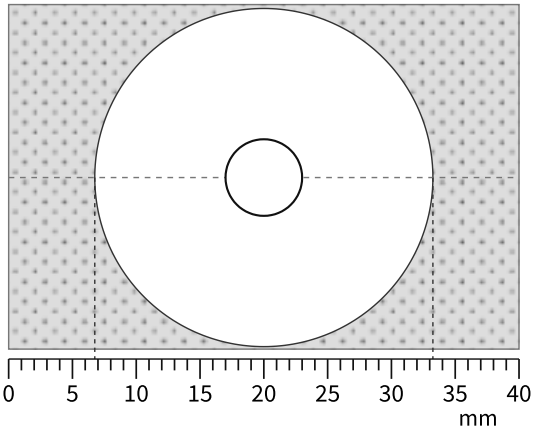
X



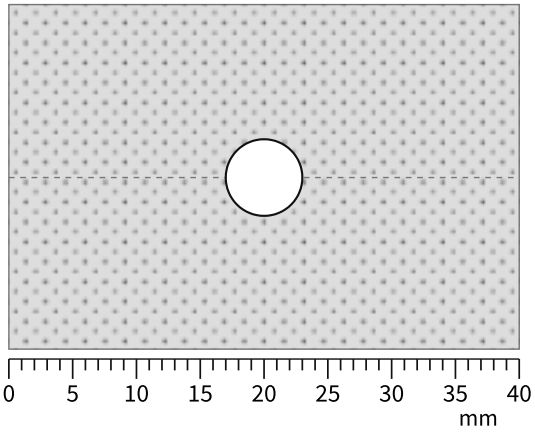
Y



Z



C (control)



Sample	Visible zone outside the disc?	Total clear-zone（清晰區） diameter (mm)
X	☐ Yes ☐ No	
Y	☐ Yes ☐ No	
Z	☐ Yes ☐ No	
C (control)	☐ Yes ☐ No	

Analyse and conclude

Use your three readings in B: mean = sum of three readings ÷ 3, to one decimal place.

Plate	X (mm)	Y (mm)	Z (mm)	C (control) (mm)
1				
2				
3				

Mean for X (mm) _____ Mean for Y (mm) _____

Mean for Z (mm) _____ Mean for C (control) (mm) _____

1. What does the appearance of a clear zone（清晰區） indicate?

○ A. All bacteria in the zone have died.
○ B. Under these test conditions, there is no visible bacterial growth in the zone, indicating inhibition（抑制）.
○ C. Bacteria in the zone resist every antibiotic（抗生素）.
○ D. The sample has eliminated viruses on the agar plate（瓊脂板）.

2. What does the diameter of a clear zone（清晰區） represent?

○ A. The exact number of bacteria remaining on the plate.
○ B. Its effectiveness in patients; the largest diameter identifies the best treatment.
○ C. The extent of the area with no visible bacterial growth under these conditions; diffusion and other factors also affect the diameter.
○ D. The antibiotic（抗生素） dose a patient should receive.

3. Draw a conclusion from your results:

Under these simulated conditions, samples （X ／ Y ／ Z ／ X and Y ／ X and Z ／ Y and Z ／ X, Y and Z ／ No samples） show a visible zone outside the disc.

Ranking of mean total clear-zone（清晰區） diameter under these conditions:

（X ／ Y ／ Z ／ C (control)） ＞ （X ／ Y ／ Z ／ C (control)） ＞ （X ／ Y ／ Z ／ C (control) ／ Not applicable） ＞ （X ／ Y ／ Z ／ C (control) ／ Not applicable）

Rank only samples with an external zone; select “Not applicable” for unused places. This is not a clinical effectiveness or treatment ranking.

4. Why do experiments need to be repeated?

○ A. To ensure that every experiment produces exactly identical absolute values, with no variation whatsoever.
○ B. To prolong the experiment so scientists have more to describe when writing their reports.
○ C. To reduce random error and assess the reproducibility（可重複性） of the results and the reliability（可靠性） of the data.
○ D. To make biological samples rapidly mutate and develop entirely new traits through repeated procedures.

5. If repeated results differ greatly, what should you do, and why?

○ A. Delete results that disagree with the original hypothesis so the results match.
○ B. Keep only the largest reading because it best represents inhibition（抑制）.
○ C. Calculate the mean immediately because averaging removes all procedural problems.
○ D. First check spreading, disc positions and plate conditions, then repeat as needed to identify causes of variation and improve the comparison.

6. When comparing the clear zones（清晰區） around C (control) and X, Y and Z, what is the main role of C?

○ A. Ensure that all discs have clear zones（清晰區） of the same diameter.
○ B. Identify the best antibiotic（抗生素） for a patient.
○ C. Provide an antibiotic（抗生素）-free comparison to help assess whether clear-zone（清晰區） formation is associated with the antibiotic（抗生素） sample.
○ D. Increase the amount of sample in the X, Y and Z discs to increase the accuracy（準確度） of the experimental results.

This model uses no clinical susceptibility（對抗生素的敏感性） breakpoints and cannot be generalised to all MRSA（耐甲氧西林金黃葡萄球菌）.

Concept application: selection of resistance（抗藥性）

Choose a growth-inhibiting（抑制） sample and cite readings. If a few bacteria already resist that antibiotic（抗生素）, how might continued use affect its effectiveness? Explain through survival, reproduction（繁殖）, inheritance（遺傳） and changing proportions.

Evidence to cite

Refer to your own twelve readings and means in B3–6. Use one effective sample; a short causal explanation is enough.

Complete B yourself and preserve your first answers before reading C. The examples in A are not the results for X/Y/Z/C.

Name: _____ Class: _____ No.: _____

Check your measurements and analysis

Distribute only after B is completed and first answers preserved. Check, then correct in another colour without erasing the original.

Plate	Sample	Visible zone outside the disc?	Total clear-zone（清晰區） diameter (mm)
1	X	Yes	18.7
	Y	No	0.0
	Z	Yes	26.1
	C (control)	No	0.0
2	X	Yes	18.7
	Y	No	0.0
	Z	Yes	26.3
	C (control)	No	0.0
3	X	Yes	18.7
	Y	No	0.0
	Z	Yes	26.5
	C (control)	No	0.0

Mean total diameter calculation	X	Y	Z	C (control)
mm	18.7	0.0	26.3	0.0

Reference values are not a reason to overwrite valid readings. Ruler readings may differ within ±1 mm; check means against your own readings.

1. What does the appearance of a clear zone（清晰區） indicate?	B. Under these test conditions, there is no visible bacterial growth in the zone, indicating inhibition（抑制）.
2. What does the diameter of a clear zone（清晰區） represent?	C. The extent of the area with no visible bacterial growth under these conditions; diffusion and other factors also affect the diameter.
4. Why do experiments need to be repeated?	C. To reduce random error and assess the reproducibility（可重複性） of the results and the reliability（可靠性） of the data.
5. If repeated results differ greatly, what should you do, and why?	D. First check spreading, disc positions and plate conditions, then repeat as needed to identify causes of variation and improve the comparison.
6. When comparing the clear zones（清晰區） around C (control) and X, Y and Z, what is the main role of C?	C. Provide an antibiotic（抗生素）-free comparison to help assess whether clear-zone（清晰區） formation is associated with the antibiotic（抗生素） sample.

3. Ranking of mean total clear-zone（清晰區） diameter under these conditions: Z ＞ X ＞ Not applicable ＞ Not applicable。

Samples with external zones: X、Z。

Check reasoning and correct

Distribute only after B is completed and first answers preserved. Check, then correct in another colour without erasing the original.

Your initial observations

A clear zone（清晰區） with no visible bacterial growth is present near Penicillium（青黴菌）(a fungus). This is an observation; its cause needs testing.

Independent variable	Sample in the disc
Dependent variable	Visible clear zone（清晰區） and its total diameter
Controlled variables	Disc size ／ Composition of the agar plate（瓊脂板） ／ Type of bacteria used ／ Initial bacterial distribution ／ Preset sample preparation
What are the assumptions of this investigation?	Bacteria are initially distributed evenly on each agar plate（瓊脂板）. ／ Equipment does not introduce contamination. ／ All plates use the same simulated observation conditions.

Use a matching carrier-only disc without antibiotic（抗生素） to check carrier/disc effects, with other conditions the same. Call it a water control only if every carrier is water.

Hypothesis evaluation/application

B is testable because it compares a sample/carrier under shared conditions using zone presence and diameter. Accept a reasoned conditional hypothesis for U and another bacterium, without requiring an outcome guess.

Labels, spacing and errors

Use X/Y/Z/C once each, in any order. Space discs apart and away from the edge to avoid overlap and difficult interpretation. A risks contamination; B changes bacteria and makes the comparison unfair; C is appropriate.

Repeat-plan comparison

Plan B uses new independently prepared plates to compare variation. Remeasuring one plate estimates reading variation, not independent experimental variation. Three is a classroom choice; balance resources, time and useful evidence.

Procedure and errors

Order: R → U → Q → T → S → P. A: check even spreading. B: prepare a new independent plate rather than call a second reading a new result. Retain differing data and investigate causes.

Concept application

For example, X has readings 18.7, 18.7, 18.7 mm and an external zone. Pre-existing resistant（具抗藥性的） bacteria are more likely to survive and reproduce（繁殖）; traits can be inherited（遺傳）. Their proportion may rise, so effectiveness may fall without a complete takeover. Accept Z with its own readings. The experiment does not prove resistance（抗藥性） arose.

My correction and reason (if originally correct, state the check used)

Correct the evidence or causal link in your concept application.

Teacher only; do not distribute D or the complete pack to students

Teaching arrangements and study conditions

1. The experimental group uses IBL + VL. The comparison group has no pre-lesson lecture and does not operate the VL; it learns through the WS. Make A available from the start: explanation → example → guided practice → independent application → checking/correction.
2. Both groups use 15–20 minutes, preferably the same 20-minute limit: concepts 3, examples 2, guided practice 4, independent application 8, checking/correction 3 minutes. If using 15 minutes, use the same limit for both groups; speed is not an ability score.
3. Teachers may briefly clarify terms, diagram-reading instructions and response requirements already written in the WS, without a separate concept lecture before self-study. Examples do not use X/Y/Z/C outcomes. Do not reveal B answers before completion. Record additional prompts and actual completion times.
4. B1–2 are guided; B3–5 contain twelve independent measurements; B6 covers means and six analysis categories; B7 applies resistance（抗藥性） selection. Accept short sentences or key terms. Two lines provide space; they need not be filled.
5. After B is completed, preserve the first answers before distributing C for corrections. Only then provide answers and focused feedback. Do not give answers to unfinished students. Record unfinished items and pilot feasibility of 15–20 minutes; software tests do not establish actual learning duration.
6. All diagrams and rulers use existing fixed seeds 1001, 1002 and 1003 with generateResult. Individual VL seeds differ; do not claim all learner values are identical. 30°C and 24 hours are simulation conditions. No physical MRSA（耐甲氧西林金黃葡萄球菌） culture is required.
7. Check external-zone measurements within ±1 mm of the model. Absence requires No and exactly 0. Check means against the student’s own readings (±0.1 mm), avoiding a second penalty for one measurement error.
8. The 38-mark WS check is specific to this task version, not directly comparable with the VL’s 23 marks; paper ordering/reading is not practical performance. Use common SPSAI/CKT assessments for study outcomes; researchers must verify the formal instruments and their validity.
9. This redesign follows the researcher’s 14 October 2026 comparison condition, replacing the earlier same-item, inquiry-before-teaching arrangement. Use the alignment table to update study documents. Preserve approved bilingual vocabulary; do not change learner VL, records, cloud collection or Excel.

Ordering and diagram reading do not demonstrate virtual practical skill. WS marks are not equivalent to VL prediction, operation or completion scores. Evaluate group outcomes using separate common assessments.

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Check your measurements and analysis

Distribute only after B is completed and first answers preserved. Check, then correct in another colour without erasing the original.

Plate	Sample	Visible zone outside the disc?	Total clear-zone（清晰區） diameter (mm)
1	X	Yes	18.7
	Y	No	0.0
	Z	Yes	26.1
	C (control)	No	0.0
2	X	Yes	18.7
	Y	No	0.0
	Z	Yes	26.3
	C (control)	No	0.0
3	X	Yes	18.7
	Y	No	0.0
	Z	Yes	26.5
	C (control)	No	0.0

Mean total diameter calculation	X	Y	Z	C (control)
mm	18.7	0.0	26.3	0.0

Reference values are not a reason to overwrite valid readings. Ruler readings may differ within ±1 mm; check means against your own readings.

1. What does the appearance of a clear zone（清晰區） indicate?	B. Under these test conditions, there is no visible bacterial growth in the zone, indicating inhibition（抑制）.
2. What does the diameter of a clear zone（清晰區） represent?	C. The extent of the area with no visible bacterial growth under these conditions; diffusion and other factors also affect the diameter.
4. Why do experiments need to be repeated?	C. To reduce random error and assess the reproducibility（可重複性） of the results and the reliability（可靠性） of the data.
5. If repeated results differ greatly, what should you do, and why?	D. First check spreading, disc positions and plate conditions, then repeat as needed to identify causes of variation and improve the comparison.
6. When comparing the clear zones（清晰區） around C (control) and X, Y and Z, what is the main role of C?	C. Provide an antibiotic（抗生素）-free comparison to help assess whether clear-zone（清晰區） formation is associated with the antibiotic（抗生素） sample.

3. Ranking of mean total clear-zone（清晰區） diameter under these conditions: Z ＞ X ＞ Not applicable ＞ Not applicable。

Samples with external zones: X、Z。

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Check reasoning and correct

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Your initial observations

A clear zone（清晰區） with no visible bacterial growth is present near Penicillium（青黴菌）(a fungus). This is an observation; its cause needs testing.

Independent variable	Sample in the disc
Dependent variable	Visible clear zone（清晰區） and its total diameter
Controlled variables	Disc size ／ Composition of the agar plate（瓊脂板） ／ Type of bacteria used ／ Initial bacterial distribution ／ Preset sample preparation
What are the assumptions of this investigation?	Bacteria are initially distributed evenly on each agar plate（瓊脂板）. ／ Equipment does not introduce contamination. ／ All plates use the same simulated observation conditions.

Use a matching carrier-only disc without antibiotic（抗生素） to check carrier/disc effects, with other conditions the same. Call it a water control only if every carrier is water.

Hypothesis evaluation/application

B is testable because it compares a sample/carrier under shared conditions using zone presence and diameter. Accept a reasoned conditional hypothesis for U and another bacterium, without requiring an outcome guess.

Labels, spacing and errors

Use X/Y/Z/C once each, in any order. Space discs apart and away from the edge to avoid overlap and difficult interpretation. A risks contamination; B changes bacteria and makes the comparison unfair; C is appropriate.

Repeat-plan comparison

Plan B uses new independently prepared plates to compare variation. Remeasuring one plate estimates reading variation, not independent experimental variation. Three is a classroom choice; balance resources, time and useful evidence.

Procedure and errors

Order: R → U → Q → T → S → P. A: check even spreading. B: prepare a new independent plate rather than call a second reading a new result. Retain differing data and investigate causes.

Concept application

For example, X has readings 18.7, 18.7, 18.7 mm and an external zone. Pre-existing resistant（具抗藥性的） bacteria are more likely to survive and reproduce（繁殖）; traits can be inherited（遺傳）. Their proportion may rise, so effectiveness may fall without a complete takeover. Accept Z with its own readings. The experiment does not prove resistance（抗藥性） arose.

My correction and reason (if originally correct, state the check used)

Correct the evidence or causal link in your concept application.

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WS task check: 38 marks (not a cross-group outcome score)

Ordering and diagram reading do not demonstrate virtual practical skill. WS marks are not equivalent to VL prediction, operation or completion scores. Evaluate group outcomes using separate common assessments.

Item	Marks	Criterion
Initial observation	2	Describe reduced visible growth/clear area near the fungus and separate observation from causal inference; 1 each.
Hypothesis evaluation	1	Select B and identify comparison conditions and an observable outcome; 1 for a reasoned judgment.
Hypothesis in another context	2	Testable comparison of U/carrier for another bacterium: 1; observable zone/diameter outcome: 1. Do not score outcome guessing.
Identify the variables	4	Independent: 1; dependent: 1; five correct controls: 0.4 each, with 0 for inappropriate extra control choices.
What are the assumptions of this investigation?	1	Only even distribution, no added contamination and shared conditions; missing/additional choices: 0.
Control explanation	2	Same carrier without antibiotic（抗生素）: 1; compare carrier/other factors under shared conditions: 1. Not necessarily a water control.
Complete labels and check fair comparison	4	Four labels once each: 1; spacing/edge reasoning: 1; identify A contamination and B unfair comparison: 1 each. No art or operation score.

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WS task check: 38 marks (not a cross-group outcome score)

Ordering and diagram reading do not demonstrate virtual practical skill. WS marks are not equivalent to VL prediction, operation or completion scores. Evaluate group outcomes using separate common assessments.

Item	Marks	Criterion
Independent repeats and feasibility	3	Plan B and independent new plates: 1; variation comparison: 1; three is not universal, with time/resource feasibility: 1.
Diagram reading and procedure ordering (not practical performance)	4	R→U→Q→T→S→P: 2 (1/3 per correct position); A spreading problem and B non-independent result: 1 each. Procedure understanding only.
Measure and record	4	Twelve readings at 1/3 each: correct zone classification plus ±1 mm; absence requires No and exactly 0.
Mean total diameter calculation	2	Four means at 0.5 each, based on the student’ s own three readings, within ±0.1 mm.
Analyse the results and draw a conclusion	6	Five MCs at 1 each: B, C, C, D, C. Item 3: samples 0.5, ranking by own means 0.5.
Resistance（抗藥性）- selection application	3	Effective sample/readings: 1; differential survival, reproduction（繁殖） and inheritance（遺傳）: 1; proportion may rise/effectiveness may decline: 1. No complete takeover required.

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Item alignment: original, treatment, new item, shared goal and rationale

VL original	Treatment	New WS item	Shared goal	Rationale
Your initial observations	Retain / static image	Refer to the historical image in A. Describe visible bacterial growth near the fungus, separating observation from an explanation of its cause.	Distinguish observation from inference	Retain historical evidence without VL operation.
Which samples do you predict will have an area without visible bacterial growth around the disc?	Rewrite: evaluate a hypothesis	Compare two claims: A, “This sample must kill all MRSA（耐甲氧西林金黃葡萄球菌）”; B, “Under the same conditions, the total clear-zone（清晰區） diameter around a disc with sample M is greater than around the carrier control.” Circle the more testable claim (A/B), and explain why clear-zone（清晰區） data can test it.	Testable hypothesis	Concepts are taught first; do not pretend to make an initial prediction.
Which sample do you predict will have the largest clear zone（清晰區） around its disc?	Rewrite: another context	Another context: compare unknown sample U with the same carrier control using a different bacterium. Write a hypothesis testable with external-zone presence and total diameter; do not make a new prediction about X/Y/Z.	Transfer testable hypotheses	Use unknown U and another bacterium instead of re-predicting seen outcomes.
Your reason (Hint: antibiotics（抗生素） can diffuse through the agar plate（瓊脂板）.)	Integrate with evaluation/application	Compare two claims: A, “This sample must kill all MRSA（耐甲氧西林金黃葡萄球菌）”; B, “Under the same conditions, the total clear-zone（清晰區） diameter around a disc with sample M is greater than around the carrier control.” Circle the more testable claim (A/B), and explain why clear-zone（清晰區） data can test it.	Explain using observable evidence	Evaluate testability and retain reasoning practice.
Independent variable	Retain, with an earlier example	Independent variable	Identify the changed factor	Use a disc-size example, then classify sample contents in X/Y/Z.
Dependent variable	Retain	Dependent variable	Identify observed/measured outcomes	Still classify zone presence and total diameter.
Controlled variables	Retain	Controlled variables	Fair-comparison conditions	Retain the same options and control goals.
What are the assumptions of this investigation?	Retain with definition	Investigation assumptions (choose all that apply)	Identify assumptions	Do not assume absence of visible growth proves complete killing.

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Item alignment: original, treatment, new item, shared goal and rationale

VL original	Treatment	New WS item	Shared goal	Rationale
If you want to find out whether antibiotics（抗生素） affect bacterial growth, how would you design a control group?	Rewrite: explain the given control	What is C? Explain its comparison baseline and why the remaining conditions must be the same.	Control comparison function	C has the same carrier without antibiotic（抗生素）, not necessarily water.
Draw and label your plates ／ Setup labels / written design	Rewrite: labels, spacing and error identification	Label the four discs X, Y, Z and C, once each. Explain why discs need space between them and from the edge. Label order does not affect credit.	Setup and fair comparison	Retain position and contamination judgments without autonomous operation.
There are 4 test samples: X, Y, Z and the control. How many tests do you think each sample needs in total? (Include the first test.)	Rewrite: compare given plans	Plan A: remeasure the same plate three times. Plan B: independently prepare and observe three new plates once each. Which provides independent repeats, and how do they help check variation?	Independent repeats and variation	Distinguish new plates from remeasuring one zone.
Why would you plan it this way?	Rewrite: feasibility reasoning	This lesson uses three plates. Why is “three” not the only correct number for every study? What feasibility limits matter when comparing three with more independent plates?	Evidence quality and feasibility	Three is a classroom arrangement, not a universal standard.
Experiment and record ／ Procedure	Static: before/after images, ordering and errors	These cards describe a simulated procedure. Order the letters logically; prepare three plates independently before the shared observation. Answering does not demonstrate sterilisation（滅菌／消毒）, spreading or incubation（培養） performance.	Procedure understanding and evidence interpretation	Not evidence of sterilisation（滅菌／消毒）, spreading or incubation（培養） performance.
Visible zone outside the disc? ／ Measure and record the total clear-zone（清晰區） diameter (mm).	Retain: twelve static rulers	Measure and record the total clear-zone（清晰區） diameter (mm).	Measure and classify	Use the same seeds/model; do not preprint B answers.
Mean total diameter calculation	Retain with a separate worked example	Mean total diameter calculation	Calculate from own readings	Four three-reading means; avoid repeated measurement penalties.
What does the appearance of a clear zone（清晰區） indicate?	Same item	What does the appearance of a clear zone（清晰區） indicate?	Meaning of clear-zone（清晰區） evidence	Independent interpretation after teaching; no complete-killing inference.

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Item alignment: original, treatment, new item, shared goal and rationale

VL original	Treatment	New WS item	Shared goal	Rationale
What does the diameter of a clear zone（清晰區） represent?	Same item	What does the diameter of a clear zone（清晰區） represent?	Diameter meaning and limits	Do not infer patient treatment from zone size.
Draw a conclusion from your results: ／ Comparing the mean total clear-zone （清晰區） diameters in this test, their inhibitory（抑制） effects rank as follows:	Equivalent rewrite: mean-diameter order	Ranking of mean total clear-zone（清晰區） diameter under these conditions:	Draw bounded conclusions from data	Retain sample selection without calling diameter clinical effectiveness.
Why do experiments need to be repeated?	Same item	Why do experiments need to be repeated?	Purpose of repeats	Retain random error, reproducibility（可重複性） and reliability（可靠性）.
If repeated results differ greatly, what should you do, and why?	Same item	If repeated results differ greatly, what should you do, and why?	Evaluate variation and methods	Check causes; do not delete data for disagreement.
When comparing the clear zones（清晰區） around C (control) and X, Y and Z, what is the main role of C?	Same item	When comparing the clear zones（清晰區） around C (control) and X, Y and Z, what is the main role of C?	Role of the control	Retain comparison against non-antibiotic（抗生素） factors.
Choose an antibiotic（抗生素） sample from your experimental results that can inhibit（抑制） bacterial growth, and cite readings to support your judgement. Suppose this bacterial population initially contains a small number of bacteria resistant（具抗藥性的） to that antibiotic（抗生素）. How might its effectiveness change with continued use? Use the learning points to explain how bacterial survival, reproduction（繁殖） and inheritance（遺傳） of resistance traits（抗藥性特徵） cause this change.	Equivalent concise concept application	Choose a growth-inhibiting（抑制） sample and cite readings. If a few bacteria already resist that antibiotic（抗生素）, how might continued use affect its effectiveness? Explain through survival, reproduction（繁殖）, inheritance（遺傳） and changing proportions.	Data, survival, reproduction（繁殖） and inheritance（遺傳）	The original already applies resistance（抗藥性） selection; no hypothesis-revision task is needed.
Inquiry submitted ／ Learning points	Reorder: concepts in A, checking in C	Distribute only after B is completed and first answers preserved. Check, then correct in another colour without erasing the original.	Self-study and correction	Teach before practice; preserve first answers and do not score completion as ability.